

Pham Hung Quoc Viet

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OBJECTIVE

3rd-year Information Systems student at University of Information Technology – VNU-HCM (UIT) with practical experience in Machine Learning and AI model development. Experienced using LLM APIs (OpenAI, Gemini) and currently researching AI agent architectures and business applications. Seeking an **AI Research Intern** position at Abstract Agency to integrate AI APIs into products, develop AI-driven automation workflows, and contribute to research that reduces operational time and cost.

EDUCATION

University of Information Technology – VNU-HCM (UIT) 2023 – 2027 (expected)
Bachelor of Information Systems GPA: 8.06/10
Relevant Coursework: Machine Learning, Data Mining, Information Systems Analysis & Design, Database Management Systems, Web Development

SKILLS

AI & LLM	OpenAI API, Gemini API, Prompt Engineering, AI Agent concepts (self-studied)
Machine Learning	Scikit-learn, XGBoost, TensorFlow, SHAP, Optuna, SMOTE
Programming	Python (Pandas, NumPy, Matplotlib), SQL, PHP
Data & Automation	PySpark, Kafka, Docker, PostgreSQL, Streamlit, Git/GitHub
BI & Databases	Power BI, SSIS, SSAS, SQL Server, PostgreSQL
Languages	IELTS 6.5 (2022) – proficient English technical documentation reader

PROJECTS

Bank Customer Churn Prediction & AI Explainability 2026
Python, Scikit-learn, TensorFlow, XGBoost, LightGBM, SMOTE, SHAP, Optuna | [GitHub](#)

- AI Research & Model Selection:** Independently researched and systematically benchmarked 7+ ML algorithms (XGBoost, LightGBM, TensorFlow DNN, Random Forest) with Optuna hyperparameter tuning – selecting the optimal model for a real business problem, mirroring a structured AI research workflow.
- Data Engineering & Preprocessing:** Applied Winsorization, feature encoding, and SMOTE oversampling to a 10k-record dataset (6,400 → 10,192 balanced samples), boosting model recall by improving training data quality on imbalanced classes.
- Explainable AI (XAI) for Business:** Leveraged SHAP and LIME to surface the top churn drivers (active membership, product count); ran Jaccard similarity analysis (0.596) to cross-validate findings – translating black-box AI decisions into actionable insights for non-technical stakeholders.
- Research Documentation & Proposal:** Produced structured benchmark reports with model-vs-model comparisons; proposed optimization strategies (threshold tuning, ensemble stacking) as next steps – reflecting the periodic reporting requirement of AI research roles.

NYC Taxi — Automated Real-time AI Anomaly Detection System 2026
PySpark, Kafka, Docker, PostgreSQL, Streamlit, Pandas, Scikit-learn | [GitHub](#)

- End-to-end AI Workflow Design:** Architected a fully automated AI pipeline from real-time data ingestion (Kafka at 200 msg/s) through stream processing (PySpark, 14 engineered features) to anomaly scoring and reporting – reducing manual monitoring to zero.
- Production AI Model Integration:** Embedded an Isolation Forest + MinCovDet ensemble (strict-voting to minimize false positives) directly into a live streaming pipeline, enabling autonomous real-time decisions without human intervention.
- Self-adapting AI Agent Principle:** Implemented automated model retraining triggered every 25 data batches, allowing the system to continuously adapt to concept drift – a core behavior of production-grade AI agents.
- Operational Cost Reduction:** Delivered a Streamlit monitoring dashboard with auto-alerts, replacing manual shift-by-shift reviews and demonstrating measurable time savings through AI-driven workflow automation.